

2024



Snappet Boosts Uptime and Scalability, Improves Service to Teachers and Students Using AWS

Education technology firm [Snappet](#) created a microservices infrastructure on Amazon Web Services (AWS) to scale faster and better meet the needs of schools, teachers, and students. The Netherlands-based firm launched in 2012 on AWS with the vision of making “excellent teaching easier” and bringing educational success to children. But the company needed a more agile, responsive, and robust environment to meet growing demand for its digital learning technology.

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Zero downtime

for teachers and students

Tri

capa

Increased

insight and operational efficiency



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Overview

Educational technology (EdTech) firm Snappet is passionate about teaching. The company's digital learning system, launched in 2012, is used by over 3,000 elementary schools in the Netherlands, Belgium, Spain, and the US. It aims to make teaching the core curriculum easy, driving better educational outcomes through connected learning experiences. Snappet's 200 staff—former teachers, education specialists, developers, and subject-matter experts—work directly with schools to develop interactive teaching solutions.

With its technology playing such an important role in students' education, it's vital for Snappet to ensure uninterrupted access to resources at all times. But as demand for its platform increased, it became increasingly challenging to innovate quickly and maintain reliability, so the company looked to modernize its infrastructure. Snappet is an Amazon Web Services (AWS) customer, and it built a microservices environment using serverless services, helping it to scale fast and maintain service continuity for schools. With the help of AWS, it also incorporated compliance with the strictest privacy regulations. Now, with a more agile architecture, developers have greater capacity to innovate, teachers and students have seamless access to materials, and the business is prepared for continued growth.



Opportunity | Finding an Agile Way to Meet Growing Global Demand

Snappet's driving goal has always been simple: improve teaching and learning through technology. And its e-learning platform—which allows teachers to plan lessons, tailor learning, and monitor performance—has proved a great success, quickly going from prototype to launching across the Netherlands, and then Europe and the US.

Each time the company needed to onboard a new school to the platform, Snappet's IT team would onboard new schools in centralized databases. However, with growing demand, this monolithic architecture was struggling to keep up. Plus, Snappet was now running applications not just for teachers and students, but for administrators and support staff too. Availability had become an issue, and Snappet's engineering teams were increasingly interrupted by the need for manual work to keep the platform running.

The team decided to scale horizontally, clustering schools across multiple database systems. As demand grew, performance and availability issues, it impeded innovation. "We needed to increase development speed, so we



added teams,” says Klaas Waslander, chief technical officer of Snappet. “We added lots of features, but we quickly saw that we couldn’t work with multiple teams in one central web application—it wouldn’t be possible to keep innovating quickly due to the monolithic architecture with centralized database systems. We needed a way to let multiple teams develop in parallel.”

Solution | Creating a Scalable Environment for EdTech Innovation Using AWS

Looking for a new approach, Snappet engaged with AWS solution architects for advice. “It helped us review our IT environment as well as our organizational architecture,” says Waslander. “We knew we had all the tools we needed in the AWS Cloud to help our developers work in a more efficient and innovative way.”

With the objective of tripling development capacity, Snappet started restructuring its environment around microservices. “We created multi-accounts on AWS and set up DevOps teams that could use whatever tools were best suited to their needs,” says Waslander.

Instead of the centralized database systems that the company launched with, Snappet has migrated to a microservices-based architecture with multiple databases. These include [Amazon Aurora](#), which provides unparalleled high performance and availability at global scale with full MySQL and PostgreSQL compatibility, and [Amazon DynamoDB](#), a serverless, NoSQL, fully managed database service with single-digit millisecond response times. Some development teams choose to go serverless with [AWS Lambda](#) to run code without thinking about servers or clusters. Others use [Amazon Elastic Container Service](#) (Amazon ECS), which can run highly secure, reliable, and scalable containers.

According to Waslander, the new setup is transforming the way Snappet operates. “Using AWS, we’re able to scale when and where needed,” he says. “Developers can innovate, test, and deploy without getting in each other’s way. And platform security is improved because data and services are clearly segmented. This also simplifies disaster recovery because if a service fails it’s easy to launch new instances to restore functionality.”

He adds, “On the business side, the multi-account setup is also giving us better insight into what services are using resources, allowing us to apply FinOps principles to ensure efficient use of resources.”



And although Snappet hasn't looked back with its new architecture, it continues to benefit from advice and assistance from AWS Partners. "We have a lot of smart, independently minded people here and it seems we did a lot right," says Waslander, after the AWS Partner helped the team conduct an [AWS Well-Architected](#) review to check its infrastructure against best practices. "It was great to confirm we're still on the right track. Plus, we got some good advice to improve the organization overall."

Outcome | New Approach Drives Better Experience for Schools, Teachers, Students—and Developers

Reliability issues are now a thing of the past, supporting Snappet's mission of making excellent teaching easier. Teachers can deliver engaging lessons without interruption and students have a better learning experience with access to all the resources and tools they need. Thanks to help from AWS, Snappet also knows that it complies with strict privacy regulations. And, having tripled development capacity thanks to the new DevOps model, features are more frequently deployed, constantly improving the functionality of the e-learning platform for users.

The company's developers are happier too. "They can run their projects as they see fit, with the resources they need," says Waslander. "They don't have to submit proposals to me or an architecture committee and wait. They can just get on with delivering secure and privacy-first EdTech innovations."

About Snappet

Snappet, founded in 2012, develops technology to help children learn. Its digital platform, created and maintained by 200 education specialists and developers, is used by 3,000 elementary schools in Europe and the US for core curricula teaching.

AWS Services Used

Amazon Aurora

Amazon Aurora provides built-in security, continuous backups, serverless compute, up to 15 read replicas, automated multi-Region replication, and integrations with other AWS services.

[Learn more »](#)

Amazon DynamoDB

Amazon DynamoDB is a fully managed, serverless, key-value NoSQL database designed to deliver consistent performance applications at any scale.



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Amazon Lambda

AWS Lambda is a serverless, event-driven compute service that lets you run code for virtually any type of application or backend service without provisioning or managing servers.

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Amazon ECS

Amazon Elastic Container Service (Amazon ECS) is a fully managed container orchestration service that makes it easy for you to deploy, manage, and scale containerized applications.

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The University of Applied Sciences and Arts Northwestern (FHNW) is one of Switzerland’s largest universities. In collaboration with interactive museum for analog and digital technology Enter Technikwelt Solothurn, FHNW embarked on a generative artificial intelligence (AI) project using Amazon Web Services (AWS). With managed service Amazon SageMaker, says research associate Claudio Paonessa,



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